# FDAM: A Frequency-Direction-Aware Multi-Task Framework for Pavement Crack Grid Localization

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## Abstract

*Pavement crack grid localization remains challenging for thin and tiny cracks due to weak visual cues and background interference. Box detection often misses these cracks, while supervised semantic segmentation faces limited pixel-wise annotations and high labeling costs. In this paper, we propose a Frequency-Direction-Aware Multi-Task framework (FDAM) with box detection and grid classification branches. The core grid classification branch comprises Frequency-Domain Enhancement (FDE) and Directional Feature Refinement (DFR). FDE adaptively weights spectral bands and fuses reconstructed features with spatial features. DFR captures thin crack patterns through multi-scale strip convolutions, direction-sensitive modulation, and feature recalibration. Sparse positive cells and ambiguous local predictions motivate adaptive grid supervision. Adaptive Importance Reweighting emphasizes uncertain predictions and local label transitions, with focal supervision for all cells. Our contribution is to couple the box detection and grid classification branches through shared feature learning for pavement crack grid localization. We validate FDAM on our RoadDistress2026 dataset of thin and tiny cracks and further evaluate it on public datasets Crack500 and DeepCrack. The code is available at* https://github.com/Lab-PANbin/FDAM.

## 1. Introduction

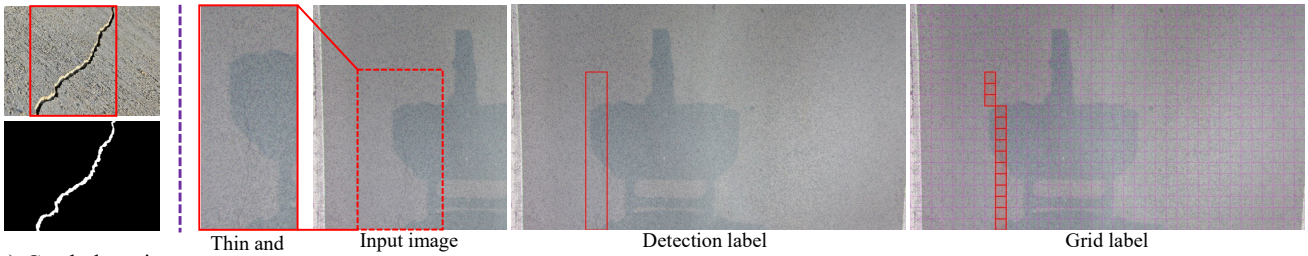
Pavement crack detection identifies surface cracks and their spatial distribution for road condition assessment [2, 35]. We focus on pavement crack grid localization for thin and tiny cracks. We use thin to describe narrow apparent widths, whereas tiny describes limited spatial extent relative to the image. These image-space characteristics, together with pavement textures and shadows, make reliable localization challenging [8].

Existing methods represent cracks using bounding boxes or pixel-wise masks. Transformer-based detectors predict categories and instance extents through set prediction [16, 17, 37, 52]. Convolutional segmentation aggregates multi-scale features [1, 46, 55], while lightweight and attention-based designs refine crack details [25, 50, 54]. Transformers [28, 31, 45] and state space models [9, 11, 32, 53] extend contextual modeling. Multiscale and hybrid architectures integrate local detail with broader context [10, 12, 43, 44], with further refinement through structural modeling and feature mixing [13, 41, 49, 56]. Spectral processing provides complementary features for restoration and segmentation [4, 20, 47, 51]. Directional models encode oriented cracks [3], while strip convolutions capture elongated objects in remote sensing [48].

However, weak crack cues, costly annotations, and imbalanced supervision still challenge thin and tiny crack localization. Bounding boxes can contain substantial background, while fully supervised segmentation requires precise pixel-wise masks. Weakly supervised methods reduce annotation demands [14, 15, 18, 21] but still need to distinguish weak cracks from background. Although tailored losses and focal loss address foreground–background imbalance [26, 29, 39], sparse positive grid cells and ambiguous label transitions remain challenging. Reliable grid localization therefore requires stronger crack representations and effective supervision across the image.

Grid occupancy provides this local representation and has an engineering basis in pavement assessment. China's *Highway Performance Assessment Standard* (JTG 5210–2018) classifies longitudinal and transverse asphalt pavement cracks as low-severity when the main crack width does not exceed 3 mm. Its automated assessment of the Pavement Surface Condition Index uses occupied $0.1\ \mathrm{m} \times 0.1\ \mathrm{m}$ grids to estimate distress area [34]. Inspired by this practice and prior joint box–grid learning [22], we combine instance localization and local occupancy prediction for thin and tiny cracks. Our grids operate in image coordinates without a prescribed physical size. Fig. 1 illustrates the task and its

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(a) Crack detection and segmentation

(b) Pavement crack grid localization

Figure 1. Illustration of pavement crack localization tasks and labels. (a) Crack detection and segmentation. Detection and pixel-wise segmentation labels are shown for a wide crack. (b) Pavement crack grid localization for thin and tiny cracks. From left to right: input image and crack detail, detection label, and grid label. The full-image grid defines regular classification units, and cells with highlighted boundaries indicate crack presence.

complementary labels.

To address this challenge, we propose FDAM, a Frequency–Direction-Aware Multi-Task framework that couples box detection and grid classification through shared feature learning. A shared backbone and hybrid encoder support a transformer-based detection branch [16] and the core grid classification branch. Within the grid branch, Frequency-Domain Enhancement (FDE) adaptively weights spectral bands, while Directional Feature Refinement (DFR) captures elongated crack structures. Adaptive Importance Reweighting emphasizes uncertain predictions and local grid-label transitions while retaining focal supervision over all cells. During inference, class-aware box–grid association filters grid predictions using detection categories and spatial overlap. Experiments on RoadDistress2026, Crack500, and DeepCrack evaluate the framework and its components.

Our main contributions are summarized as follows:

- We develop a multi-task framework for pavement crack grid localization of thin and tiny cracks through shared box and grid feature learning.
- We design a grid classification branch combining adaptive spectral weighting and directional feature refinement to strengthen weak crack representations.
- We introduce adaptive importance reweighting for grid focal supervision, emphasizing uncertain predictions and local label transitions while retaining supervision over all cells.

## 2. Related Work

### 2.1. Crack detection and segmentation

Automated pavement analysis spans distress evaluation [6], crack localization [7], quantification [36], and learned distress recognition [5]. Transformer-based detectors predict object categories and boxes through set prediction [16, 17, 37, 52]. For pixel-wise crack prediction, convolutional networks aggregate hierarchical and multiscale features [1, 46, 55]. Lightweight designs and attention-based integration improve efficiency and feature representation [23, 25, 50, 54]. Transformers capture contextual dependencies through attention [28, 31, 45]. Selective state space modeling [9] also supports long-range interactions in recent crack segmentation networks [11, 32, 53].

Preserving spatial detail remains essential for weak crack representations. High-resolution multiscale features support semantic segmentation [10], while crack-specific architectures integrate local detail with broader context [12, 43, 44]. Other methods explore convolution–transformer integration [41], convolutional and state space features [13], topology-aware modeling [56], and global–local feature mixing [49]. These approaches advance dense crack prediction, but fully supervised segmentation requires precise pixel-wise annotations.

### 2.2. Frequency and directional feature modeling

Spectral and directional representations provide complementary cues for recovering weak structures. Frequency-domain processing has been explored in image inpainting [4] and crack segmentation under shadows [51]. Related work investigates high-frequency refinement for weakly supervised segmentation [47] and frequency-aware decoding for thin cracks [20]. Directional modeling represents cracks as oriented sub-cracks [3], while strip convolutions capture elongated objects in remote sensing images [48]. These studies motivate combining spectral and directional cues in our grid classification branch. Because cracks and pavement textures can share frequency components, we combine input-dependent spectral weighting with directional refinement.

### 2.3. Weak supervision and loss reweighting

Annotation requirements and foreground sparsity motivate alternative supervision strategies. Weakly supervised crack analysis uses coarse annotations or generated labels to reduce dependence on precise masks [14, 15, 18, 21]. Loss

design has also been investigated for imbalanced crack segmentation [26, 39]. Focal loss reduces the contribution of well-classified examples [29]. Our supervision operates on category-specific grid occupancy. Adaptive Importance Reweighting uses prediction uncertainty and local grid-label transitions to weight focal loss while supervising all entries. These transitions describe coarse occupancy changes without requiring pixel-wise crack boundaries.

## 3. Methodology

We formulate pavement crack grid localization as joint instance detection and category-wise grid occupancy prediction. FDAM couples shared feature learning with spectral enhancement, directional refinement, and adaptive grid supervision.

### 3.1. Overall framework

Given an input image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$, a backbone and hybrid encoder produce three shared feature maps $\{\mathbf{F}_l\}_{l=1}^{3}$. The two branches predict

$$\mathcal{D}_{\text{raw}} = \{(\hat{\mathbf{p}}_i, \hat{\mathbf{b}}_i)\}_{i=1}^{N_q}, \mathbf{Z} \in \mathbb{R}^{C_g\times H_g\times W_g}, \tag{1}$$

where $N_q$ is the query count, $\hat{\mathbf{p}}_i$ and $\hat{\mathbf{b}}_i$ are category scores and normalized boxes, and $C_g$ categories share an $H_g \times W_g$ grid.

As shown in Fig. 2, both branches learn from shared features while retaining task-specific prediction heads. The detection branch follows DEIMv2 [16], using a transformer decoder and Hungarian matching for set prediction. We retain its classification, box regression, and auxiliary losses in $\mathcal{L}_{\text{det}}$. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{grid}}\mathcal{L}_{\text{grid}}, \tag{2}$$

where $\lambda_{\text{grid}}$ balances the objectives. Both losses update the shared backbone and encoder.

### 3.2. Grid classification branch

Pointwise projections and bilinear resizing align the shared features to $C$ channels and the coarsest encoder resolution $(H_g, W_g)$. Concatenation and convolution produce the fused feature $\mathbf{F}_g$. Fig. 3 shows the subsequent enhancement and prediction stages. Detailed operators are given in Supplementary Sec. A.

#### 3.2.1. Frequency-Domain Enhancement

Thin and tiny cracks produce spatial variations across multiple frequency bands, which may also contain pavement textures and noise. FDE therefore learns input-dependent spectral weights. Let $\mathcal{F}_r$ denote the orthonormal real Fourier transform. Four disjoint radial masks $M_k$ receive weights

$$\begin{aligned} \boldsymbol{\omega} &= \text{Softmax}(\phi_b(\text{GAP}(\mathbf{F}_g))), \\ M_b &= \sum_{k=1}^{4} \omega_k M_k, M_f = M_b \odot (1 + \chi_f M_d), \end{aligned} \tag{3}$$

where GAP denotes global average pooling, $M_d \in [0, 1]$ is a learned axis-modulation mask, and $\chi_f \in \{0, 1\}$ specifies whether it is enabled.

The masks are shared across channels. Spectral reconstruction and gated fusion give

$$\begin{aligned} \mathbf{F}_f &= \mathcal{F}_r^{-1}[(1 + \lambda_f M_f) \odot \mathcal{F}_r(\mathbf{F}_g)], \\ \mathbf{G}_f &= \sigma(\phi_g(\mathbf{F}_g \| \mathbf{F}_f)), \\ \mathbf{F}_{\text{e}} &= \mathbf{F}_g + \mathbf{G}_f \odot (\mathbf{F}_f - \mathbf{F}_g), \end{aligned} \tag{4}$$

where $\lambda_f$ is learned, $\sigma$ is sigmoid, $\|$ denotes channel concatenation, and $\odot$ denotes element-wise multiplication.

The spatial gate is broadcast across channels. Since $\|M_f\|_\infty \le 2$, the correction satisfies

$$\|\mathbf{F}_{\text{e}} - \mathbf{F}_g\|_F \le 2|\lambda_f|\,\|\mathbf{F}_g\|_F. \tag{5}$$

Thus, $|\lambda_f|$ controls the relative correction magnitude for each input, and $\lambda_f = 0$ recovers the input. Mask construction and the derivation appear in Supplementary Sec. B.

#### 3.2.2. Directional Feature Refinement

DFR captures elongated crack patterns through multiscale strip convolutions, directional modulation, and feature recalibration. Each stage applies parallel $1 \times k$ and $k \times 1$ convolutions with $k \in \{3, 7, 11\}$, followed by concatenation, normalization, and activation to obtain $\mathbf{A}$. Axis-wise pooling produces attention maps $\mathbf{T}_h, \mathbf{T}_v$, while global pooling predicts softmax weights $a_h, a_v$. The directional update is

$$\begin{aligned} \mathbf{A}_d &= \mathbf{A} \odot (a_h \mathbf{T}_h + a_v \mathbf{T}_v), \\ \mathbf{F}_{\text{out}} &= \mathbf{F}_{\text{in}} + \text{GN}(\text{Conv}_{1\times1}(\mathbf{A}_d)). \end{aligned} \tag{6}$$

where GN denotes group normalization.

Applying $N_b$ independently parameterized stages to $\mathbf{F}_{\text{e}}$ yields $\mathbf{F}_r$. Two complementary spatial maps reweight equal channel groups, followed by channel recalibration:

$$\begin{aligned} \mathbf{F}_s &= (\mathbf{F}_{r,1} \odot \mathbf{S}_1) \| (\mathbf{F}_{r,2} \odot \mathbf{S}_2), \\ \mathbf{F}_d &= \mathbf{F}_s \odot \sigma(\psi_c(\text{GAP}(\mathbf{F}_s))). \end{aligned} \tag{7}$$

The maps satisfy $\mathbf{S}_1 + \mathbf{S}_2 = 1$ at each location. The channel groups are learned partitions without prescribed orientations. Spatial dropout and a pointwise classifier produce $\mathbf{Z}$. Supplementary Sec. A.2 specifies the attention, recalibration, and prediction operators.

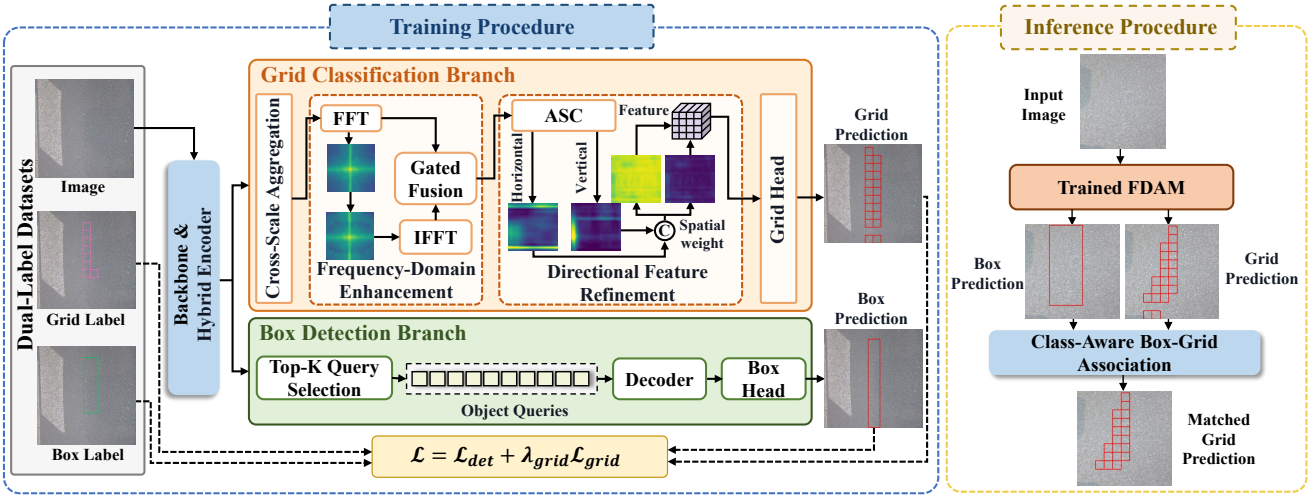



Figure 2. Overview of FDAM for pavement crack grid localization. The training procedure jointly optimizes a box detection branch and a grid classification branch through shared feature learning. The inference procedure applies class-aware box-grid association to refine grid predictions using detection categories and spatial overlap.

### 3.3. Adaptive importance reweighting

Grid supervision emphasizes uncertain predictions and local label transitions while retaining supervision over every category-cell entry. Grid annotations are stored separately from detection boxes. Their transformed centers and eight offsets are projected onto the prediction grid to form $\mathbf{Y} \in \{0,1\}^{C_g \times H_g \times W_g}$. Categories are encoded independently, allowing multiple labels per cell. Supplementary Sec. C.1 details target construction.

For training, let $\bar{\mathbf{Z}} = \mathrm{clip}(\mathbf{Z}, -20, 20)$ and $\mathbf{P} = \sigma(\bar{\mathbf{Z}})$. Define

$$\begin{aligned} \mathbf{U} &= 1 - |2\mathbf{P} - 1|, \\ \mathbf{T} &= \mathrm{Dilate}_{k_t}(\mathbf{Y}) - \mathrm{Erode}_{k_t}(\mathbf{Y}), \\ \mathbf{W}_{\mathrm{AIR}} &= \mathrm{StopGrad}(1 + \lambda_u \mathbf{U} + \lambda_t \mathbf{T}). \end{aligned} \tag{8}$$

Uncertainty peaks at $P = 0.5$, and $\mathbf{T}$ marks coarse grid-label transitions rather than pixel-wise crack boundaries. StopGrad prevents differentiation through the weighting rule. The grid objective uses binary focal loss [29]:

$$\mathcal{L}_{\mathrm{grid}} = \frac{\sum_j W_{\mathrm{AIR},j}\, \ell_{\mathrm{focal}}(P_j, Y_j)}{\max(1, N_+)}, N_+ = \sum_j Y_j, \tag{9}$$

where $j = (b, c, u, v)$ indexes the local mini-batch.

With $\lambda_u = \lambda_t = 1$, the weights satisfy $1 \le W_j \le 3$. Relative to the unit-weight objective $\mathcal{L}_0$,

$$\frac{\partial \mathcal{L}_{\mathrm{grid}}}{\partial \bar{Z}_j} = W_{\mathrm{AIR},j} \frac{\partial \mathcal{L}_0}{\partial \bar{Z}_j}. \tag{10}$$

The weights preserve each clipped-logit's gradient sign while scaling its magnitude by a bounded positive factor.

The focal formulation, hyperparameters, and derivation are provided in Supplementary Sec. C.2.

### 3.4. Inference procedure

A single forward pass produces both outputs. Decoded detections with scores $s^d > \tau_d$ form $\mathcal{D}$. Each grid cell selects its highest-scoring category from $\sigma(\mathbf{Z})$ and is retained as a candidate when $s^g \ge \tau_g$. After mapping both outputs to original-image coordinates and a common category space, association retains

$$\mathcal{G}^* = \left\{ g \in \mathcal{G} \;\middle|\; \begin{array}{l} \exists d \in \mathcal{D}:\ \hat{c}^g = \hat{c}^d, \\ |\mathbf{b}^g \cap \mathbf{b}^d| > 0 \end{array} \right\}. \tag{11}$$

Association only filters grid candidates and preserves their scores. If $\mathcal{D} = \varnothing$, then $\mathcal{G}^* = \varnothing$. Grid-only inference returns the unassociated set $\mathcal{G}$. Box average precision uses decoded detections independently of the association threshold. Supplementary Sec. D and Algorithm 1 give the complete procedure.

## 4. Experiments

We evaluate box detection and grid classification on RoadDistress2026, Crack500, and DeepCrack, followed by component and task ablations.

### 4.1. Datasets and Annotation Preparation

**RoadDistress2026.** Our dataset contains 13,000 images at $4656 \times 2657$ pixels: 9,100 for training, 2,600 for validation, and 1,300 for testing. Longitudinal cracks (LC) and transverse cracks (TC) are labeled on a $20 \times 37$ grid. Positive cells carry category and instance labels; each detection box encloses one instance's cells.

**Public Benchmarks.** Crack500 contains 500 mask-labeled images [46], processed into 1,896 training, 348 validation,

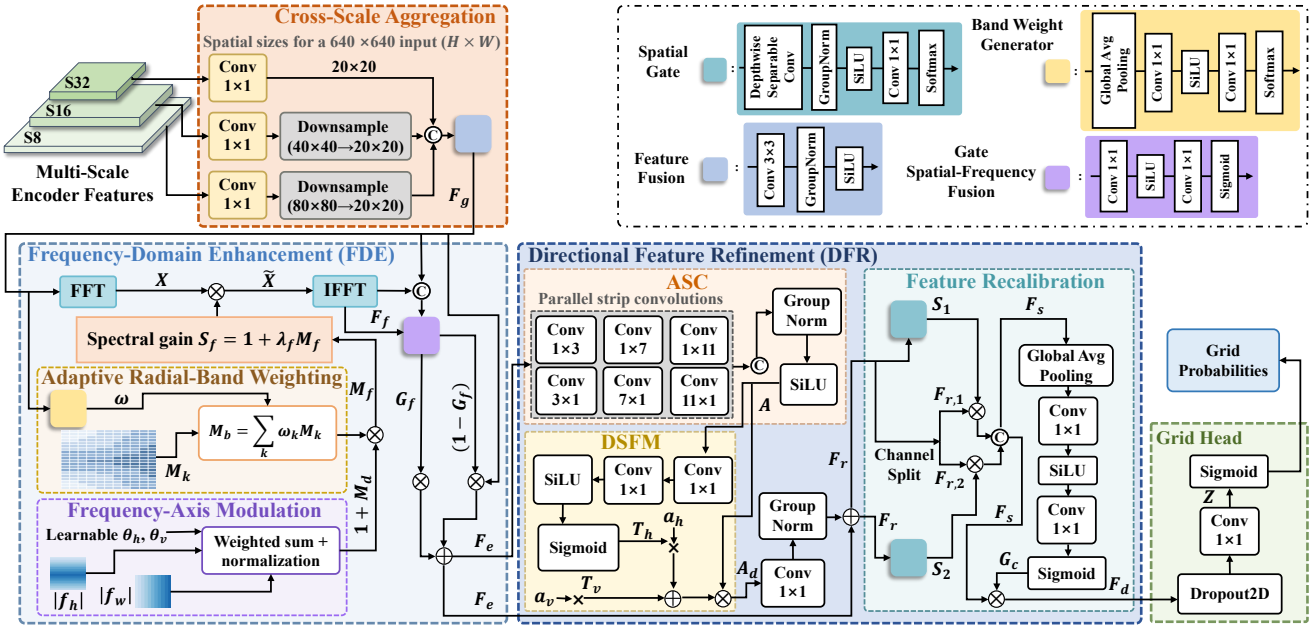



Figure 3. Grid classification branch with Frequency-Domain Enhancement and Directional Feature Refinement. Spatial dimensions illustrate a 640 × 640 input.

and 1,124 test samples. DeepCrack contains 537 images of diverse surfaces [55], split into 375/53/109 images for training/validation/testing.

For both datasets, boxes enclose 8-connected mask components with at least three foreground pixels. Grid annotations are derived separately using 20 × 20-pixel reference cells on a 640 × 640 plane, scaled to the original image size and rounded to integer pixels. These annotation cells differ from the network's 20 × 20 output grid, whose cells cover 32 × 32 input pixels. Annotation regions map to the output grid through the nine-point projection in Section 3.3, with $\delta_g = 0.125$. FDAM uses the derived boxes and grid regions. Segmentation baselines use dense masks.

## 4.2. Experimental Setup

### 4.2.1. Training Settings

FDAM uses PyTorch 2.2.2 on an Intel Xeon E5-2680 workstation with two 24GB NVIDIA RTX 3090 GPUs. AdamW uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, warm-up, and a flat-cosine schedule.

Backbones use pretrained DINOv3 representations. DeepCrack and its ablations share COCO-pretrained DEIMv2 initialization and seed zero. Training applies ImageNet normalization, photometric and geometric augmentation, scheduled Mosaic/MixUp, and multiscale resizing. The grid head uses four frequency bands and dropout 0.1. Grid loss parameters are $\alpha_f = 0.5$ and $\gamma_f = 1.5$. AIR uses $\lambda_u = \lambda_t = 1$ and $k_t = 3$.

### 4.2.2. Model Selection and Inference

Validation-selected checkpoints and thresholds are frozen for testing with exponential-moving-average weights. Joint and detection-only models are selected by box AP. Grid-only uses foreground grid F1. Both joint outputs use the same checkpoint. On public datasets, thresholds maximize validation foreground grid F1, including for segmentation baselines. Joint models use $\tau_g = 0.45$ and $\tau_d = 0.30$. DeepCrack Grid-only uses $\tau_g = 0.50$.

Joint inference retains grid cells overlapping same-category detections with positive area. Grid-only uses raw grids. Box AP is evaluated independently of grid association. RoadDistress2026 uses $\tau_d = \tau_g = 0.5$ for quantitative joint inference. All three datasets retain grid candidates with $s^g \geq \tau_g$ and detection candidates with $s^d > \tau_d$.

### 4.2.3. Evaluation Protocol

Comparisons include eight box detectors, five segmentation methods, and GCBD [22] under identical splits. Box detection uses COCO AP over IoU thresholds 0.50:0.05:0.95, $AP_{50}$, and $AP_{75}$. RoadDistress2026 reports category-wise grid precision, recall, and confidence-ranked cell AP after box association, without an IoU sweep. On public datasets, FDAM probabilities are thresholded directly on its output grid. Precision, recall, F1, and IoU use the confusion matrix aggregated over all test cells. Overall metrics average background and crack equally. Crack Only denotes foreground metrics. All accuracy values are percentages.

Table 1. Box detection on Crack500 and DeepCrack. All methods use the same mask-derived box annotations. The best and second-best results are highlighted in **bold** and underlined, respectively.

| Method | Crack500 | | | | | DeepCrack | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Params (M) | GFLOPs | AP | $AP_{50}$ | $AP_{75}$ | Params (M) | GFLOPs | AP | $AP_{50}$ | $AP_{75}$ |
| YOLOv12 [40] | 26 | 99 | 18.8 | 33.8 | 18.2 | 26 | 99 | 32.8 | 52.0 | 32.6 |
| YOLO26 [38] | 25 | 96 | 18.0 | 33.1 | 17.1 | 25 | 96 | 33.2 | 52.3 | 36.3 |
| RT-DETR [52] | 36 | 100 | 19.8 | 34.1 | 19.6 | 36 | 100 | 35.8 | 52.0 | 36.3 |
| RT-DETRv2 [33] | 36 | 100 | 19.9 | 33.9 | 20.0 | 36 | 100 | 30.2 | 44.7 | 30.6 |
| RT-DETRv3 [42] | 31 | 92 | 18.6 | 31.7 | 18.5 | 31 | 92 | 26.7 | 39.5 | 28.7 |
| RT-DETRv4 [27] | 31 | 91 | 20.1 | 34.8 | 20.0 | 31 | 91 | 25.6 | 40.6 | 24.5 |
| DEIM [17] | 31 | 92 | 20.4 | 35.0 | 20.3 | 31 | 92 | 36.9 | 54.7 | 37.7 |
| DEIMv2 [16] | 10 | 25 | 21.3 | 35.4 | 21.0 | 18 | 52 | 37.6 | 55.5 | 37.3 |
| **FDAM (Ours)** | 12 | 29 | **21.8** | **36.6** | **21.8** | 22 | 56 | **39.8** | **57.3** | **40.7** |

Table 2. Box detection on RoadDistress2026. FDAM uses only the detection branch in this comparison. LC and TC denote longitudinal and transverse cracks. The best and second-best results are highlighted in **bold** and underlined, respectively.

| Method | Params (M) | GFLOPs | LC | | | TC | | | All | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | AP | $AP_{50}$ | $AP_{75}$ | AP | $AP_{50}$ | $AP_{75}$ | AP | $AP_{50}$ | $AP_{75}$ |
| YOLOv12 [40] | 26 | 281 | 15.6 | 38.5 | 8.8 | 16.5 | 40.4 | 11.0 | 16.1 | 39.5 | 9.9 |
| YOLO26 [38] | 25 | 248 | 15.1 | 34.4 | 11.1 | 15.9 | 38.1 | 10.5 | 15.5 | 36.3 | 10.8 |
| RT-DETR [52] | 36 | 254 | 15.4 | 38.7 | 8.8 | 15.9 | 42.9 | 7.2 | 15.7 | 40.8 | 8.0 |
| RT-DETRv2 [33] | 36 | 254 | 15.0 | 37.1 | 10.3 | 16.5 | 42.8 | 9.1 | 15.8 | 40.0 | 9.7 |
| RT-DETRv3 [42] | 43 | 157 | 15.5 | 39.1 | 8.8 | 15.3 | 39.2 | 9.8 | 15.4 | 39.2 | 9.3 |
| RT-DETRv4 [27] | 31 | 224 | 19.9 | 50.2 | 12.7 | 19.1 | 50.0 | 11.2 | 19.5 | 50.1 | 12.0 |
| DEIM [17] | 31 | 230 | 19.6 | 47.4 | 12.8 | 18.6 | 47.2 | 11.4 | 19.1 | 47.3 | 12.1 |
| DEIMv2 [16] | 32 | 242 | 19.9 | 48.8 | 12.6 | 19.1 | 50.0 | 11.7 | 19.5 | 49.4 | 12.2 |
| **FDAM (Ours)** | 35 | 248 | **20.2** | **51.1** | **13.5** | **19.8** | **51.1** | **12.2** | **20.0** | **51.1** | **12.9** |

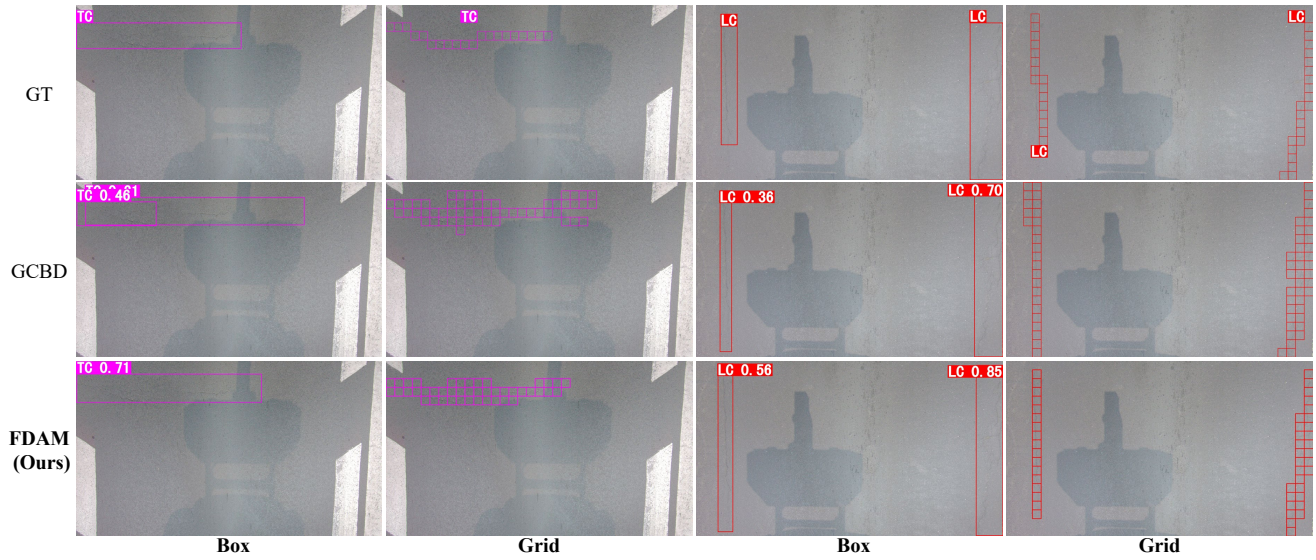



Figure 4. Qualitative comparison of GCBD and the joint FDAM model on RoadDistress2026 for box detection and grid localization.

#### 4.2.4. Input Resolution and Complexity

The public-dataset comparisons in Table 1 use $640 \times 640$ inputs. RoadDistress2026 comparisons in Tables 2 and 3

Table 3. Joint box-grid comparison on RoadDistress2026. FDAM uses the multitask model with both detection and grid branches. All denotes the average over LC and TC. Grid AP is class-wise average precision without an IoU sweep. Best results are in **bold**.

| Task | Method | Params (M) | GFLOPs | LC | | | TC | | | All | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | AP | $AP_{50}$ | $AP_{75}$ | AP | $AP_{50}$ | $AP_{75}$ | AP | $AP_{50}$ | $AP_{75}$ |
| Box | GCBD [22] | 56 | 117 | 7.6 | 25.5 | 4.0 | 13.4 | 41.8 | 3.6 | 10.5 | 33.7 | 3.8 |
| | **FDAM (Ours)** | 35 | 248 | **17.7** | **44.5** | **12.4** | **19.6** | **46.3** | **13.0** | **18.7** | **45.4** | **12.7** |
| | | | | P | R | AP | P | R | AP | P | R | AP |
| Grid | GCBD [22] | 56 | 117 | 44.0 | 61.6 | 46.7 | 53.5 | 55.6 | 51.3 | 48.8 | 58.6 | 49.0 |
| | **FDAM (Ours)** | 35 | 248 | **59.3** | **68.8** | **65.6** | **62.0** | **69.1** | **67.3** | **60.7** | **69.0** | **66.5** |

Table 4. Grid classification results for pavement crack grid localization on Crack500 and DeepCrack under a common grid protocol. Overall metrics are macro-averaged over background and crack. Segmentation baselines use dense mask supervision, while FDAM uses derived box and grid annotations. Best and second-best results are in **bold** and underlined, respectively.

| Method | Crack500 | | | | | | | | | | DeepCrack | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Params (M) | GFLOPs | Overall | | | | Crack Only | | | | Params (M) | GFLOPs | Overall | | | | Crack Only | | | |
| | | | mP | mR | mF1 | mIoU | P | R | F1 | IoU | | | mP | mR | mF1 | mIoU | P | R | F1 | IoU |
| SimCrack [19] | 299 | 146 | 84.3 | 86.2 | 85.2 | 75.7 | 72.3 | 77.2 | 74.7 | 59.6 | 299 | 146 | 89.0 | 86.5 | 87.7 | 79.2 | 82.4 | 76.0 | 79.0 | 65.3 |
| SCSegamba [32] | 2.8 | 18 | 90.5 | 90.6 | 90.5 | 83.4 | 83.5 | 83.8 | 83.7 | 71.9 | 2.8 | 18 | 94.2 | 89.6 | 91.7 | 85.2 | 91.8 | 80.5 | 85.8 | 75.1 |
| MixerCSeg [53] | 2.5 | 3 | 89.9 | 90.3 | 90.1 | 82.8 | 82.4 | 83.5 | 82.9 | 70.9 | 2.5 | 3 | 90.6 | 89.0 | 89.8 | 82.2 | 84.7 | 80.7 | 82.6 | 70.4 |
| CCMamba [30] | 89 | 41 | 90.3 | 91.4 | 90.8 | 83.9 | 82.9 | 85.7 | 84.3 | 72.8 | 89 | 41 | **95.7** | 91.3 | 93.3 | 87.8 | **94.4** | 83.4 | 88.6 | 79.5 |
| MSFAF-Net [24] | 5.5 | 28 | 87.0 | 88.6 | 87.8 | 79.4 | 77.1 | 81.1 | 79.1 | 65.4 | 5.5 | 28 | 87.5 | 85.5 | 86.4 | 77.3 | 79.5 | 74.4 | 76.9 | 62.4 |
| **FDAM (Ours)** | 12 | 29 | **91.5** | **92.7** | **92.1** | **85.8** | **84.9** | **87.8** | **86.3** | **76.0** | 22 | 56 | 93.7 | **93.7** | **93.7** | **88.5** | 89.3 | **89.4** | **89.4** | **80.8** |

use $1024 \times 1024$ inputs. In Table 4, GFLOPs use each method's input resolution: $256 \times 256$ for SimCrack and MSFAF-Net, $512 \times 512$ for SCSegamba and MixerCSeg, and $640 \times 640$ for FDAM. CCMamba's GFLOPs cover one $384 \times 384$ sliding window, excluding additional windows per image. These values describe the stated settings rather than a common-resolution efficiency comparison.

## 4.3. Comparison with Existing Methods

### 4.3.1. Box Detection on Public Benchmarks

Table 1 shows that FDAM achieves 21.8/39.8 AP on Crack500/DeepCrack, exceeding the strongest baselines by 0.5/2.2 percentage points and leading in $AP_{50}$ and $AP_{75}$.

### 4.3.2. Box Detection on RoadDistress2026

In Table 2, the detection-only configuration reaches 20.0 AP, 51.1 $AP_{50}$, and 12.9 $AP_{75}$, leading for both LC and TC. The joint model in Table 3 performs both box detection and grid classification, achieving 18.7 box AP and 66.5 grid AP. Fig. 4 presents qualitative comparisons of the detection and grid-classification results between the two methods.

### 4.3.3. Comparison on Pavement Crack Grid Localization

Table 4 compares FDAM with segmentation methods after converting their predictions to grids. FDAM leads all eight metrics on Crack500, with mF1/mIoU of 92.1/85.8, exceeding CCMamba by 1.3/1.9 percentage points. On DeepCrack, FDAM achieves the highest mF1/mIoU of 93.7/88.5. Relative to CCMamba, higher crack recall (89.4 versus 83.4) offsets lower precision (89.3 versus 94.4), yielding higher crack F1/IoU of 89.4/80.8. Fig. 5 shows qualitative comparisons under the same grid protocol.

## 4.4. Ablation Studies

DeepCrack ablations share the backbone, split, initialization, and training settings except for the component or task removed. Both joint outputs use one validation-selected checkpoint. Table 5 reports task and component ablations with all detection and grid metrics.

Table 5. Task and component ablations on DeepCrack. Grid metrics average background and crack.

| Configuration | Box Detection | | | Grid Classification | | | |
|---|---|---|---|---|---|---|---|
| | AP | $AP_{50}$ | $AP_{75}$ | mP | mR | mF1 | mIoU |
| Detection-only | 35.5 | 51.1 | 36.8 | – | – | – | – |
| Grid-only | – | – | – | 93.8 | **94.5** | **94.2** | **89.2** |
| Joint w/o FDE | 38.4 | 54.7 | 40.4 | 93.3 | 93.6 | 93.4 | 88.0 |
| Joint w/o DFR | 38.6 | 54.1 | 38.7 | 93.3 | 93.4 | 93.4 | 87.9 |
| Joint w/o AIR | 37.8 | 54.8 | 38.7 | **95.6** | 87.4 | 90.9 | 84.0 |
| Full FDAM | **39.8** | **57.3** | **40.7** | 93.7 | 93.7 | 93.7 | 88.5 |

**Adaptive Importance Reweighting.** AIR increases box AP from 37.8 to 39.8 and grid mIoU from 84.0 to 88.5. Higher grid macro recall, 93.7 versus 87.4, offsets lower macro precision and raises mF1 from 90.9 to 93.7. AIR is

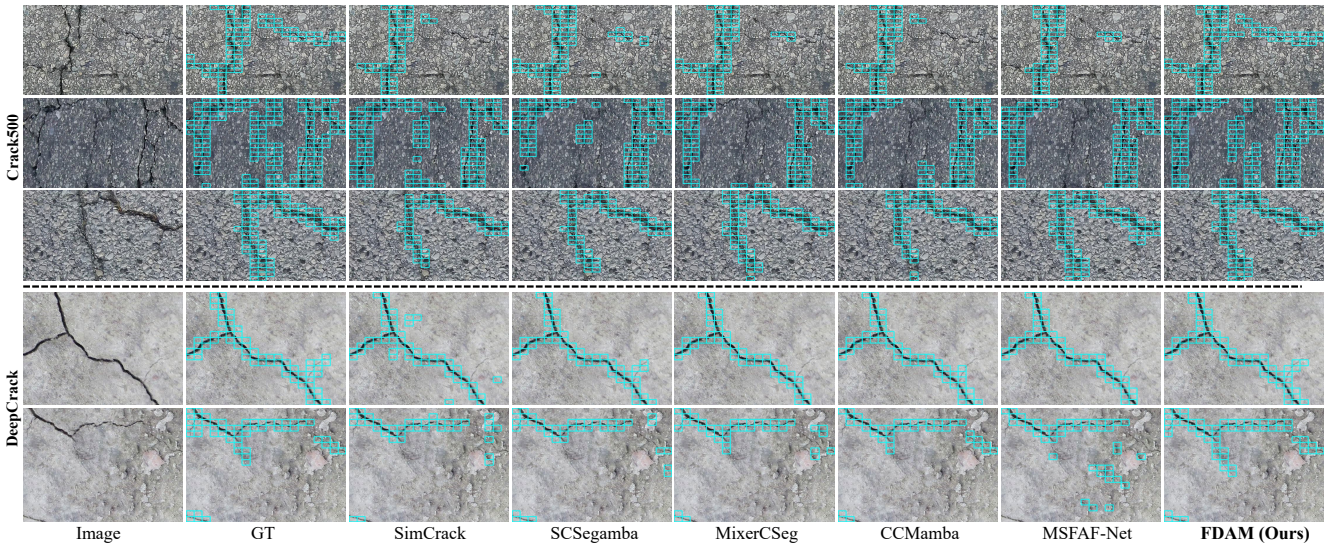



Figure 5. Qualitative comparison of grid localization on Crack500 and DeepCrack. Each row shows an input image, grid ground truth, and predictions from the compared methods. Segmentation outputs are converted to the common grid representation.

enabled in the grid-only and full joint models.

**Task branches.** The joint model exceeds detection-only AP, 39.8 versus 35.5, while grid-only achieves higher mF1/mIoU, 94.2/89.2 versus 93.7/88.5. The grid comparison does not isolate a multitask trade-off. Joint models use box-AP checkpoint selection and associated grids, whereas grid-only uses foreground-F1 selection and raw grids.

**Feature enhancement.** Removing FDE reduces box AP/grid mIoU from 39.8/88.5 to 38.4/88.0. Removing DFR reduces them to 38.6/87.9. Both components improve the measured detection and grid scores under these settings.

#### 4.4.1. Effect of the Grid Loss Weight

Table 6 compares $\lambda_{\text{grid}} \in \{0.5, 1.0, 2.0\}$ on DeepCrack with other training settings fixed. Each setting uses a checkpoint selected by validation box AP for both tasks and validation-selected grid thresholds frozen for testing. The setting $\lambda_{\text{grid}} = 1.0$ achieves the highest box AP and grid mIoU among the tested values. Grid mF1 varies from 93.5 to 93.7 across these settings.

Table 6. Effect of the grid loss weight on DeepCrack. Grid metrics are macro-averaged over background and crack. All results are percentages.

| $\lambda_{\text{grid}}$ | Box Detection | | | Grid Localization | | | |
|---|---|---|---|---|---|---|---|
| | AP | $AP_{50}$ | $AP_{75}$ | mP | mR | mF1 | mIoU |
| 0.5 | 37.7 | 55.2 | 38.1 | 93.5 | 93.5 | 93.5 | 88.1 |
| 1.0 | **39.8** | **57.3** | **40.7** | **93.7** | 93.7 | **93.7** | **88.5** |
| 2.0 | 38.2 | 54.7 | 38.6 | 93.4 | **93.9** | **93.7** | 88.4 |

## 5. Conclusion

We present FDAM for pavement crack grid localization of thin and tiny cracks. The framework couples box detection and grid classification through shared features. Spectral and directional enhancement strengthen grid representations, while adaptive importance reweighting emphasizes uncertain predictions and label transitions. Experiments on RoadDistress2026, Crack500, and DeepCrack demonstrate the effectiveness of the framework and its components. Fixed grids limit geometric detail, and missed detections can suppress valid cells during box-grid association.

## LLM Usage Statement

We used ChatGPT to assist with language refinement. It also assisted with literature searches, reference checks, code implementation , and debugging. AI assistance supported discussions of evaluation protocols, and interpretation of experimental outputs. The authors determined the final methodology and experimental design and supervised the experiments. All reported numerical results were computed by the experimental and evaluation code on the stated datasets. The authors reviewed the AI-assisted content and take full responsibility for the manuscript, code, scientific claims, and reported results.

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# FDAM: A Frequency-Direction-Aware Multi-Task Framework for Pavement Crack Grid Localization

Supplementary Material

## A. Detailed architecture

### A.1. Feature aggregation and detection

The shared encoder produces feature maps $\mathbf{F}_l \in \mathbb{R}^{C_l \times H_l \times W_l}$ at three scales. For channel width $C$ divisible by 32, pointwise projections $\phi_l$ and bilinear interpolation align them to $(H_g, W_g) = (H_3, W_3)$:

$$\begin{aligned} \mathbf{Q}_l &= \mathrm{Resize}_{H_g,W_g}(\phi_l(\mathbf{F}_l)), \\ \mathbf{F}_g &= \delta(\mathrm{GN}[\mathrm{Conv}_{3\times3}(\mathbf{Q}_1 \| \mathbf{Q}_2 \| \mathbf{Q}_3)]) \,. \end{aligned} \tag{A.1}$$

where $\delta$ is SiLU.

The detection decoder progressively updates $N_q$ queries from the shared encoder features. Prediction heads produce class scores and normalized boxes. Hungarian matching establishes one-to-one assignments to ground-truth instances, and the original detection losses are retained.

### A.2. Directional refinement and prediction

For each refinement stage, six convolution branches use $\mathcal{K} = \{3, 7, 11\}$:

$$\begin{aligned} \mathbf{A}_h &= \Big\|_{k\in\mathcal{K}} \mathrm{Conv}_{1\times k}(\mathbf{F}_{\mathrm{in}}), \\ \mathbf{A}_v &= \Big\|_{k\in\mathcal{K}} \mathrm{Conv}_{k\times 1}(\mathbf{F}_{\mathrm{in}}), \\ \mathbf{A} &= \delta(\mathrm{GN}(\mathbf{A}_h \| \mathbf{A}_v)). \end{aligned} \tag{A.2}$$

The six branches jointly output $C$ channels. Each receives $\lfloor C/6 \rfloor$ channels, with the remainder assigned to the earlier branches in concatenation order. Symmetric padding preserves spatial dimensions.

Direction-sensitive feature modulation uses

$$\begin{aligned} \mathbf{T}_h &= \sigma(\phi_h(\mathrm{Avg}_H(\mathbf{A}))), \\ \mathbf{T}_v &= \sigma(\phi_v(\mathrm{Avg}_W(\mathbf{A}))), \\ [a_h, a_v] &= \mathrm{Softmax}(\phi_d(\mathrm{GAP}(\mathbf{A}))). \end{aligned} \tag{A.3}$$

The maps have shapes $C \times 1 \times W_g$ and $C \times H_g \times 1$ and are broadcast over the missing spatial dimension. Each of $\phi_h, \phi_v$ consists of pointwise projections $C \to \max(\lfloor C/4 \rfloor, 32) \to C$, with SiLU between them. The mapping $\phi_d$ is a pointwise projection from $C$ to two outputs.

The resulting maps enter Eq. 6. Because $a_h + a_v = 1$ and the attention maps lie in $(0, 1)$, $|\mathbf{A}_d| \le |\mathbf{A}|$ element-wise before residual projection. This property does not bound the entire residual block.

After $N_b$ stages, spatial recalibration predicts

$$[\mathbf{S}_1, \mathbf{S}_2] = \mathrm{Softmax}(\psi_s(\mathbf{F}_r)). \tag{A.4}$$

The mapping $\psi_s$ applies a depthwise $3 \times 3$ convolution, a pointwise projection to $C/4$ channels, eight-group normalization, SiLU, and a two-output pointwise projection. The channel mapping $\psi_c$ in Eq. 7 uses pointwise projections $C \to C/4 \to C$ with SiLU.

All spatial and channel gates are broadcast over the corresponding dimensions when multiplied with features. The final head computes

$$\mathbf{Z} = \mathrm{Conv}^{\mathrm{cls}}_{1\times1}(\mathrm{Dropout2D}(\mathbf{F}_d)), \mathbf{P} = \sigma(\mathbf{Z}). \tag{A.5}$$

Independent sigmoid probabilities permit multiple positive categories per training cell. Inference retains only the highest-scoring category.

## B. Spectral masks and correction bound

### B.1. Frequency coordinates and masks

The orthonormal real Fourier transform stores $\mathbf{X} = \mathcal{F}_r(\mathbf{F}_g)$ with $W_f = \lfloor W_g/2 \rfloor + 1$ columns. Indices satisfy $0 \le \xi < H_g$ and $0 \le \eta < W_f$. Mask coordinates follow the implementation:

$$\begin{aligned} f_h(\xi) &= \frac{\tilde{\xi}}{H_g}, f_w(\eta) = \frac{\eta}{2W_f - 1}, \\ \tilde{\xi} &= \begin{cases} \xi, & \xi \le \lfloor H_g/2 \rfloor, \\ \xi - H_g, & \text{otherwise.} \end{cases} \end{aligned} \tag{B.1}$$

These coordinates construct masks and do not redefine the Fourier transform frequencies. In particular, the width denominator is $W_g + 1$ for even $W_g$ and $W_g$ for odd $W_g$. Writing $q(\xi, \eta) = \sqrt{f_h(\xi)^2 + f_w(\eta)^2}$ gives

$$\begin{aligned} r(\xi, \eta) &= \frac{q(\xi, \eta)}{\max_{\xi',\eta'} q(\xi', \eta') + \epsilon}, \\ M_k &= \mathbb{I}\left[\frac{k-1}{4} \le r < \frac{k}{4}\right]. \end{aligned} \tag{B.2}$$

The four radial intervals are equally spaced and disjoint. All maxima cover retained frequency bins, and $\epsilon = 10^{-8}$. The band predictor $\phi_b$ uses pointwise projections $C \to \max(\lfloor C/4 \rfloor, 16) \to 4$ with SiLU. Softmax makes the weights positive and sum to one, hence $0 \le M_b \le 1$.

When frequency-axis modulation is enabled, its mask is

$$\begin{aligned} [a_h^f, a_v^f] &= \mathrm{Softmax}([\theta_h, \theta_v]), \\ D(\xi, \eta) &= a_h^f |f_h(\xi)| + a_v^f |f_w(\eta)|, \\ M_d(\xi, \eta) &= \frac{D(\xi, \eta)}{\max_{\xi',\eta'} D(\xi', \eta') + \epsilon}. \end{aligned} \tag{B.3}$$

The learned scalars $\theta_h, \theta_v$ are initialized to 0.5 and shared across inputs and channels. The configuration flag $\chi_f$ in Eq. 3 enables or disables this term. The spectral scale $\lambda_f$ is unconstrained and initialized to 0.1. The fusion mapping $\phi_g$ uses pointwise projections $2C \to \max(\lfloor C/4 \rfloor, 16) \to 1$ with SiLU. Its sigmoid output is a spatial gate shared across channels. The real gain preserves a nonzero coefficient's phase wherever $1 + \lambda_f M_f > 0$.

The implementation evaluates the orthonormal transform using real cosine and sine matrices, including the conjugate-frequency multiplicities required by the inverse transform.

### B.2. Bound on the spectral correction

Let $\mathcal{F}$ denote the orthonormal full complex Fourier transform, and let $\widehat{M}_f$ be the conjugate-symmetric extension of $M_f$. The radial and absolute-axis masks preserve the required symmetry. The reconstruction residual is

$$\mathbf{R} = \mathbf{F}_f - \mathbf{F}_g = \lambda_f \mathcal{F}^{-1}(\widehat{M}_f \odot \mathcal{F}(\mathbf{F}_g)). \tag{B.4}$$

Parseval's identity and $\|\widehat{M}_f\|_\infty \le 1 + \chi_f$ imply

$$\begin{aligned} \|\mathbf{R}\|_F &= |\lambda_f| \, \|\widehat{M}_f \odot \mathcal{F}(\mathbf{F}_g)\|_F \\ &\le (1 + \chi_f)|\lambda_f| \, \|\mathbf{F}_g\|_F. \end{aligned} \tag{B.5}$$

Since every gate value lies in $(0, 1)$,

$$\|\mathbf{F}_e - \mathbf{F}_g\|_F = \|\mathbf{G}_f \odot \mathbf{R}\|_F \le \|\mathbf{R}\|_F \le 2|\lambda_f| \, \|\mathbf{F}_g\|_F. \tag{B.6}$$

Disabling axis modulation gives the tighter factor $|\lambda_f|$. The argument uses the full spectrum, avoiding an unweighted half-spectrum interpretation of Parseval's identity. This is a per-input correction bound rather than a bound on sensitivity to input perturbations.

## C. Grid targets and importance-weighted supervision

### C.1. Grid target construction

Grid regions are stored separately from detection boxes and undergo the same geometric transformations as the image. Each transformed region has a grid category $c$ and normalized coordinates $(c_x, c_y, w, h)$. Starting from $\mathbf{Y} = \mathbf{0}$, project nine points:

$$\begin{aligned} v &= \lfloor (c_x + d_x \delta_g w) W_g \rfloor, \\ u &= \lfloor (c_y + d_y \delta_g h) H_g \rfloor, \end{aligned} \qquad d_x, d_y \in \{-1, 0, 1\}, \tag{C.1}$$

where $\delta_g = 0.125$. Valid indices set $Y_c(u, v) = 1$, and repeated projections are merged. The implementation casts projected coordinates to integers, which agrees with the floor expression for nonnegative projected positions. Out-of-range indices are ignored. Categories remain independent, and fully annotated images without positive regions retain all-negative targets. Fig. 6 illustrates this construction and the subsequent weighting procedure.

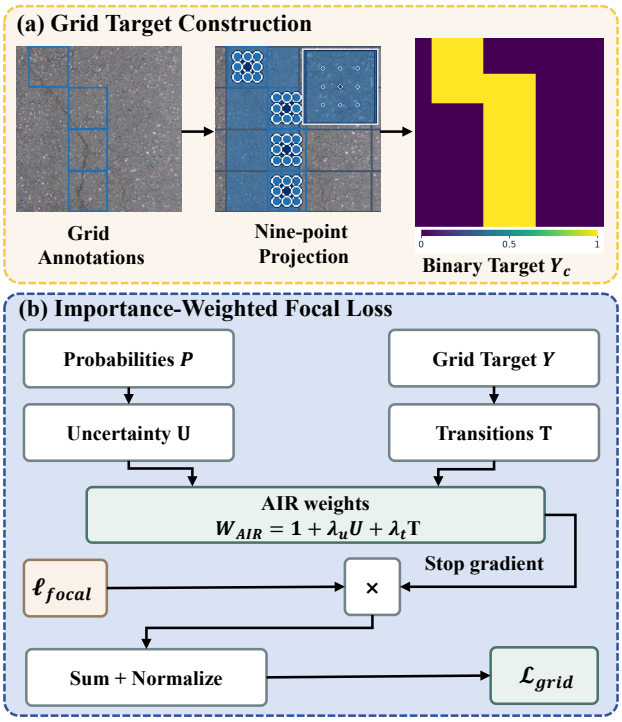



Figure 6. Grid target construction and adaptive importance reweighting. (a) Nine-point projection converts transformed grid annotations into binary category-wise targets. (b) Prediction uncertainty and target transitions determine detached weights for focal supervision over all grid entries.

### C.2. Loss formulation and gradient properties

For probability $p$ and binary target $y$, binary focal loss is

$$\begin{aligned} p_t &= yp + (1 - y)(1 - p), \\ \alpha_t &= y\alpha_f + (1 - y)(1 - \alpha_f), \\ \ell_{\text{focal}}(p, y) &= -\alpha_t (1 - p_t)^{\gamma_f} \log p_t. \end{aligned} \tag{C.2}$$

We use $\alpha_f = 0.5$, $\gamma_f = 1.5$, $\lambda_u = \lambda_t = 1$, and $k_t = 3$. Logits are clipped to $[-20, 20]$ before training probabilities and focal weights are computed. Numerically, the logarithmic term is evaluated using binary cross-entropy with logits, multiplied by the focal and class factors.

Dilation and erosion operate independently on each category using $k_t \times k_t$ neighborhoods, unit stride, and valid locations at image borders. Thus, $U_j \in [0, 1]$ and $T_j \in \{0, 1\}$. The implementation also clips importance weights to $[1, 10]$, which is inactive under the stated coefficients because $1 \le W_j \le 3$. Let $N = \max(1, N_+)$ and define

$$\mathcal{L}_0 = \frac{1}{N} \sum_j \ell_{\text{focal}}(P_j, Y_j). \tag{C.3}$$

Nonnegative focal losses yield

$$\mathcal{L}_0 \le \mathcal{L}_{\text{grid}} \le 3\mathcal{L}_0. \tag{C.4}$$

Because the weights are detached and $N$ depends only on targets,

$$\frac{\partial \mathcal{L}_{\text{grid}}}{\partial \bar{Z}_j} = \frac{W_j}{N} \frac{\partial \ell_{\text{focal}}(P_j, Y_j)}{\partial \bar{Z}_j} = W_j \frac{\partial \mathcal{L}_0}{\partial \bar{Z}_j}. \tag{C.5}$$

The resulting sign preservation applies to individual clipped logits. For raw logits, the chain rule additionally includes the clipping derivative, which is zero outside the clipping interval. The property does not imply that the full parameter gradient retains its direction or that optimization is guaranteed to converge. Every category-cell entry remains in the loss, and setting $\lambda_u = \lambda_t = 0$ recovers $\mathcal{L}_0$.

# D. Detailed inference procedure

## D.1. Decoding and category alignment

Let $\mathbf{I}_0 \in \mathbb{R}^{3\times H_0 \times W_0}$ be the original image, and let $\mathcal{T}$ denote input preprocessing. The detector's original postprocessor decodes its configured query outputs and restores boxes to original-image coordinates. Category maps $\kappa_d$ and $\kappa_g$ map detection labels and grid channels to the same semantic category space. The decoded set $\mathcal{D}_0$ is retained for box average precision. Association uses the subset with scores strictly greater than $\tau_d$.

Grid probabilities are computed as $\mathbf{P} = \sigma(\mathbf{Z})$. Each cell selects the highest-scoring category over all output channels, with ties resolved by the smallest channel index. The grid threshold uses $s^g \geq \tau_g$ on RoadDistress2026, Crack500, and DeepCrack. Under direct resizing, cell $(u, v)$ maps to

$$\mathbf{b}^g_{uv} = \left[ \frac{vW_0}{W_g}, \frac{uH_0}{H_g}, \frac{(v+1)W_0}{W_g}, \frac{(u+1)H_0}{H_g} \right]. \tag{D.1}$$

Indices are zero-based. For other supported preprocessing, the corresponding inverse geometric mapping replaces Eq. D.1.

## D.2. Association and complete algorithm

For $\mathbf{b} = [x_1, y_1, x_2, y_2]$, two boxes have positive intersection area if both intersection side lengths are positive.

Boundary touching alone does not qualify. This is equivalent to the original rule $\text{IoA}(g, d) > 0$, where $\text{IoA}(g, d) = |\mathbf{b}^g \cap \mathbf{b}^d| / (|\mathbf{b}^g| + \epsilon)$. No grid-cell expansion or positive overlap-ratio threshold is used.

Algorithm 1 specifies decoding and association for direct resizing. Grid-only mode returns unassociated predictions from the grid branch. Association never substitutes a lower-scoring category when the highest-scoring one has no matching detection. It preserves detection outputs and all retained grid scores. If there are no accepted detections, joint mode returns an empty grid set, whereas grid-only mode remains independent of detections. Box average precision is computed from $\mathcal{D}_0$ before the association confidence cutoff.

**Algorithm 1** Class-Aware Box–Grid Association

**Input:** Original image $\mathbf{I}_0$; preprocessing $\mathcal{T}$; model $\mathcal{M}$; thresholds $\tau_d, \tau_g$; category maps $\kappa_d, \kappa_g$; inference mode $m$.
**Output:** Detections $\mathcal{D}$ and grid output $\mathcal{G}_{\text{out}}$.

$\mathbf{I} \leftarrow \mathcal{T}(\mathbf{I}_0)$.
$(\mathcal{D}_{\text{raw}}, \mathbf{Z}) \leftarrow \mathcal{M}(\mathbf{I})$.
$\mathcal{D}_0 \leftarrow \text{Decode}(\mathcal{D}_{\text{raw}}, \mathcal{T}^{-1}, \kappa_d)$.
$\mathcal{D} \leftarrow \{d \in \mathcal{D}_0 \mid s^d > \tau_d\}$.
Set $\mathbf{P} \leftarrow \sigma(\mathbf{Z})$ and $\mathcal{G} \leftarrow \varnothing$.
**for** $u = 0, \ldots, H_g - 1$ **do**
  **for** $v = 0, \ldots, W_g - 1$ **do**
    $c \leftarrow \min \arg\max_j P_j(u, v)$.
    $s \leftarrow P_c(u, v)$.
    **if** $s \geq \tau_g$ **then**
      Compute $\mathbf{b}^g$ using Eq. (D.1).
      $g \leftarrow (\mathbf{b}^g, s, \kappa_g(c))$.
      $\mathcal{G} \leftarrow \mathcal{G} \cup \{g\}$.
    **end if**
  **end for**
**end for**
**if** $m =$ grid-only **then**
  **return** $\mathcal{D}, \mathcal{G}$.
**end if**
Initialize $\mathcal{G}^* \leftarrow \varnothing$.
**for** each grid candidate $g \in \mathcal{G}$ **do**
  **for** each detection $d \in \mathcal{D}$ with $\hat{c}^d = \hat{c}^g$ **do**
    $\Delta_x \leftarrow \min(x_2^g, x_2^d) - \max(x_1^g, x_1^d)$.
    $\Delta_y \leftarrow \min(y_2^g, y_2^d) - \max(y_1^g, y_1^d)$.
    **if** $\Delta_x > 0$ and $\Delta_y > 0$ **then**
      $\mathcal{G}^* \leftarrow \mathcal{G}^* \cup \{g\}$.
      **break**.
    **end if**
  **end for**
**end for**
**return** $\mathcal{D}, \mathcal{G}^*$.

# E. Experiments

We test whether FDAM improves grid localization for tinier crack instances on RoadDistress2026 and thinner crack components on Crack500 and DeepCrack. This *thin* describes narrow width in image coordinates, while *tiny* describes limited instance extent. Stratified recall identifies missed positive grid cells within each group. Overall precision and F1 check whether higher recall comes at the cost of additional false positives. A qualitative comparison further illustrates the complementary box and grid representations.

## E.1. Annotation-Defined Stratification

RoadDistress2026 uses annotated boxes because verified masks are unavailable. For a box of size $w \times h$ in an image of size $W \times H$, we define its relative extent as

$$e = \max\left(\frac{w}{W}, \frac{h}{H}\right). \tag{E.1}$$

This measures instance extent rather than crack width. For Crack500 and DeepCrack, we extract 8-connected components with at least three pixels from the original ground-truth masks. For component $C$, its relative width proxy is

$$r(C) = \frac{\mathrm{median}_{p \in \mathrm{Skel}(C)}\, 2\,\mathrm{EDT}_C(p)}{\min(H, W)}, \tag{E.2}$$

where $\mathrm{Skel}$ denotes skeletonization and $\mathrm{EDT}_C$ is the Euclidean distance to background. Masks are neither resized nor dilated before measurement. This digital proxy has discretization bias: a one-pixel-wide strip has a proxy width of two pixels. It does not measure physical width.

For each dataset, we estimate two boundaries from training annotations alone, using percentiles at $33\frac{1}{3}\%$ and $66\frac{2}{3}\%$. These boundaries, $q_1$ and $q_2$, remain fixed for test evaluation. Values $x \le q_1$, $q_1 < x \le q_2$, and $x > q_2$ define the lower, middle, and upper groups, respectively. We call them Tiny, Medium, and Large for instance extent, and Thin, Medium, and Thick for the width proxy.

Equal measurements remain in the same group, so group sizes may differ. Specifically, $r(C) \le q_1^r$ defines the thin group on each public dataset, while $e \le q_1^e$ defines the tiny group on RoadDistress2026 used to examine performance on tiny crack instances. The superscripts indicate the measurement, and the boundaries are estimated independently for each dataset. These relative groups operationalize width and extent for this analysis. They imply neither universal size thresholds nor a correspondence to physical crack width. RoadDistress2026 boxes support extent analysis only and cannot establish whether a crack is physically thin.

## E.2. Evaluation Protocol

We retain the existing project splits, checkpoints, and fixed inference settings. The test sets contain 1,300 RoadDistress2026, 1,124 Crack500, and 109 DeepCrack images. RoadDistress2026 uses direct resizing to $1024 \times 1024$ and a $32 \times 32$ grid. FDAM uses a grid threshold of $\tau_g = 0.40$ on RoadDistress2026 and $\tau_g = 0.45$ on the public datasets. GCBD uses $\tau_g = 0.35$ for unassociated grids and $\tau_g = 0.15$ for associated output. Both detection methods use $\tau_d = 0.30$. Grid scores satisfy $s_g \ge \tau_g$, whereas detection scores satisfy $s_d > \tau_d$. Association requires positive overlap with a same-category box. On Crack500, the pixel thresholds are $0.25$ for SCSegamba and $0.35$ for MixerCSeg. Both use $0.05$ on DeepCrack. Segmentation probabilities are bilinearly resized to $640 \times 640$ and thresholded. At least $1\%$ foreground occupancy marks a cell positive in the $20 \times 20$ grid. These comparisons therefore include conversion and operating-point differences. We report the recomputed outputs at these fixed settings.

Grid-group membership is assigned to source annotation regions before the original nine-point target projection. Mixed or unresolved cells, including those involving excluded small components, are omitted only from stratified recall. They remain in overall metrics. The assigned proportions of positive category-cell entries are $99.70\%$, $98.04\%$, and $98.19\%$, respectively. Within each group, recall is the fraction of ground-truth positive category-cell entries correctly predicted as positive. RoadDistress2026 averages recall over longitudinal and transverse crack categories. Public-dataset recall concerns the single crack class. Overall RoadDistress2026 scores also average these two categories. Overall precision and F1 include background false positives without averaging the background class. Box recall uses confidence-ordered, same-category matching at $\mathrm{IoU} \ge 0.50$ before grouping ground-truth instances. It measures recall at a fixed threshold, not AP.

## E.3. Stratified Results

On RoadDistress2026, associated FDAM exceeds GCBD's separately selected grid output by $5.17$ percentage points in small-extent grid recall. Fig. 7 additionally shows small-extent box recall of $43.40\%$ versus $3.77\%$. These results concern instance extent, not actual crack width. However, association reduces FDAM's overall grid F1 from $64.53\%$ to $64.24\%$ and small-extent grid recall from $46.97\%$ to $44.50\%$.

FDAM preserves more positive grid responses in the relative Thin groups of both public datasets. As shown in Fig. 8, its gains over SCSegamba are $11.45$ percentage points on Crack500 and $15.61$ on DeepCrack. These point-estimate gains exceed those in the medium and thick groups. Overall foreground F1 is also higher, although DeepCrack precision is lower than SCSegamba's. These observations do not establish statistical significance or attribute the gains to individual modules.

## E.4. Crack500 Split Sensitivity

A filename-based audit identifies five source keys shared between 36 training and 32 test images. Six test images closely match training images after a $180^\circ$ rotation, with RGB mean absolute differences of 1.25–1.43 on the 0–255 scale. These keys are inferred from filenames rather than verified acquisition metadata, and this audit is not complete content deduplication. Conservatively excluding all 32 test images leaves 1,092 images. With unchanged models and thresholds, thin-group recall remains $85.23\%$ for FDAM

Table 7. Grid recall within each annotation-defined group and overall grid performance, in percent. RoadDistress2026 groups reflect instance extent, whereas public-dataset groups reflect relative mask width. GCBD (grid) uses unassociated predictions. GCBD (box–grid) and all FDAM results use class-aware box–grid association. The grid thresholds for these output modes are fixed separately as specified in Sec. E.2.

| | Recall within each group | | | Overall grid performance | | |
|---|---|---|---|---|---|---|
| **RoadDistress2026** | | | | | | |
| Method | Tiny | Medium | Large | Precision | Recall | F1 |
| GCBD (grid) | 39.32 | 58.07 | 61.47 | 48.77 | 58.61 | 52.95 |
| GCBD (box–grid) | 6.53 | 23.24 | 31.93 | 44.33 | 26.67 | 32.96 |
| FDAM (box–grid) | 44.50 | 60.90 | 72.25 | 61.28 | 67.51 | 64.24 |
| **Crack500** | | | | | | |
| Method | Thin | Medium | Thick | Precision | Recall | F1 |
| SCSegamba | 73.72 | 84.44 | 86.03 | 83.51 | 83.84 | 83.67 |
| MixerCSeg | 72.71 | 84.79 | 85.69 | 82.31 | 83.59 | 82.95 |
| FDAM | 85.17 | 88.16 | 88.28 | 84.89 | 87.84 | 86.34 |
| **DeepCrack** | | | | | | |
| Method | Thin | Medium | Thick | Precision | Recall | F1 |
| SCSegamba | 70.84 | 81.01 | 83.29 | 91.83 | 80.50 | 85.79 |
| MixerCSeg | 67.67 | 80.53 | 84.91 | 84.61 | 80.64 | 82.58 |
| FDAM | 86.45 | 88.43 | 90.64 | 89.35 | 89.43 | 89.39 |

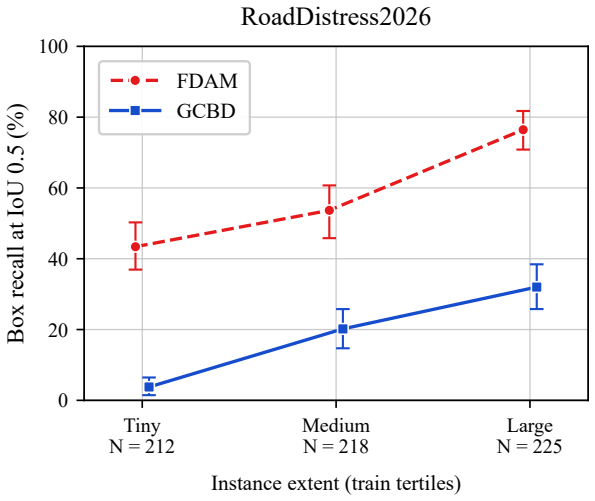



Figure 7. RoadDistress2026 box recall across relative instance extents at score $> 0.30$ and IoU $\geq 0.50$. $N$ counts annotated instances within each group. Error bars denote 95% image-bootstrap intervals.

versus 73.70% for SCSegamba, and overall foreground F1 is 86.41% versus 83.70%. The observed advantage therefore persists under this exclusion.

## E.5. Qualitative Comparison

Fig. 9 compares GCBD and FDAM on a transverse-crack example from RoadDistress2026. Box predictions describe the overall instance extent, while grid predictions represent local crack occupancy. The enlarged grid views expose differences in cell coverage that are difficult to inspect in the full images. This example illustrates the complementary spatial representations used for pavement crack grid localization. Stratified quantitative results are reported in Table 7, and broader dataset-wide comparisons are provided in the main paper.

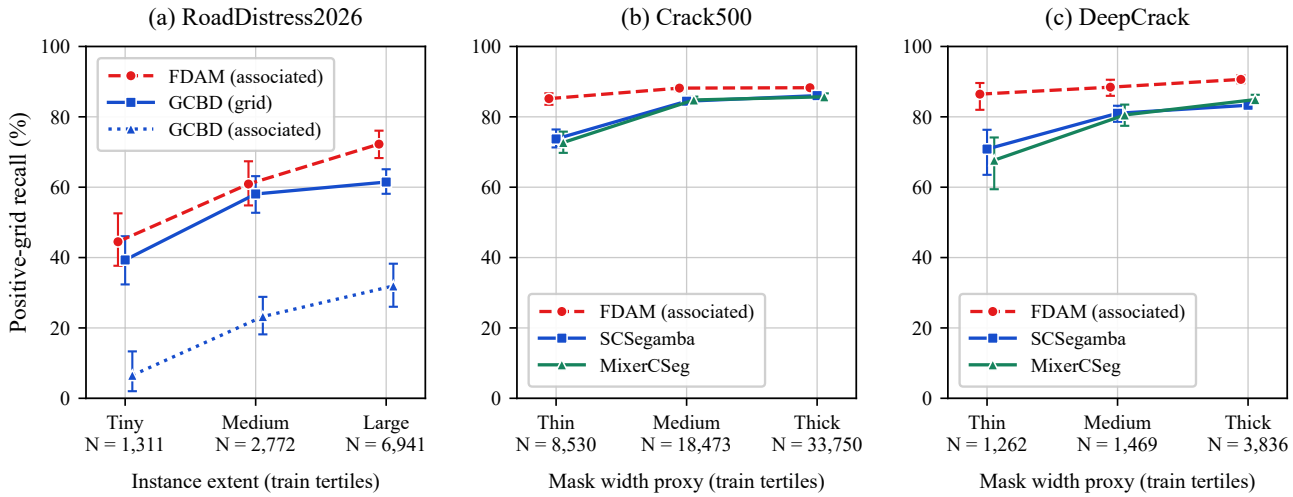



Figure 8. Grid recall across annotation-defined strata. RoadDistress2026 uses relative instance extent, whereas Crack500 and DeepCrack use the ground-truth mask width proxy. The protocols are specified in Sec. E.2. $N$ counts positive category-cell entries within each group. Error bars denote 95% image-bootstrap intervals.

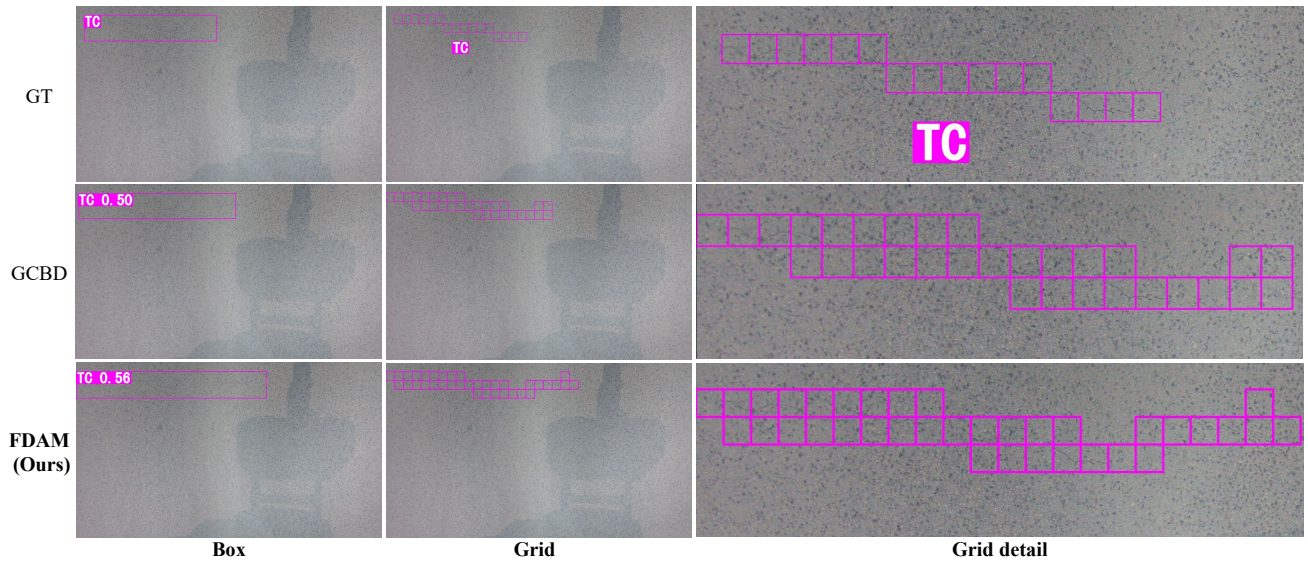



Figure 9. Qualitative comparison of pavement crack grid localization on RoadDistress2026. Rows show ground-truth (GT), GCBD predictions, and FDAM predictions. Columns show box labels or predictions, grid labels or predictions, and enlarged views of the grid results. Magenta denotes transverse cracks (TC). The enlarged views facilitate inspection of local crack occupancy.